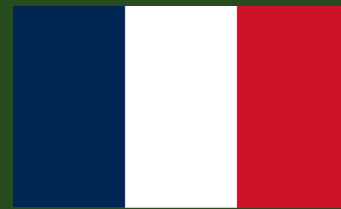


Forecasting UK Trade Route Shipping Volume Across top 3 Trading Partners - Germany, Netherlands & France



- By Group 15

AGENDA

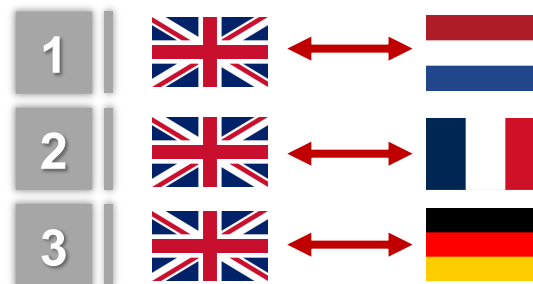
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|----------------------------------|----------------------------|
| ① Problem & Motivation | ⑥ Model |
| ② Literature Review | ⑦ Evaluation & Results |
| ③ Data Overview | ⑧ Insight & Implications |
| ④ Variables & Economic Intuition | ⑨ Limitations & Conclusion |
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Problem & Motivation

Problem Statements

- What really drives container shipping volumes between the UK and key European trade partners?
- How do freight rates, fuel costs, policy uncertainty, port congestions and disruptions affect container shipping demand?

Geographic Focus



6 Import-Export Routes

Research Beneficiaries



Motivation

- Shipping is the backbone of the UK trade economy: container shipping facilitates the majority of the UK's international goods trade
- Understanding the price sensitivity of demand: essential to determine pricing dynamics and their influence on trade flows, shipping line revenues, and competitiveness of UK-EU corridors.
- Measuring impact of geopolitical and macroeconomic shocks (COVID-19, GFC etc): events such as the GFC, Brexit and COVID-19 have exposed vulnerabilities in global supply chains
- Improving strategic planning under uncertainty supports better forecasting, capacity planning, and infrastructure investment decisions

Literature Review - Economic and Methodological Foundations

Economic Foundations

Study	Key Findings	How it Supports our Model
Short-sea shipping service design Methodology	To estimate freight demand trade and maritime transport data are used.	Supports using container volume (TEU) as a measure of shipping demand
Road freight transport decoupling (UK–Spain)	The link between freight transport demand and macroeconomic activity.	Supports using industrial production indices as macroeconomic demand controls
Maritime fuel cost competitiveness study	A major component of shipping operating expenses are the fuel costs.	Supports including bunker fuel price as a key cost-side explanatory variable
EU Blue Economy Report	The importance of maritime transport in European trade and economic activity.	Motivates forecasting shipping demand to support operational and financial planning

Methodological Foundations

Methodological Literature	Method Used	How it Supports our Model
Transport Demand Modelling	To estimate freight demand relationships linear regression is widely used.	Justifies using linear regression as an interpretable baseline model
Econometric Forecasting	When predictors are correlated, regularized regression improves stability.	Supports use of Ridge regression to improve robustness and generalization
Panel Data Econometrics	Panel data methods allow controlling for route-specific heterogeneity	Supports use of panel dataset structure with route and direction controls

References: Santos et al. (2021); Alises et al. (2014); Solakivi et al. (2022); Luo et al. (2009); Agnolucci et al. (2015); Diebold & Mariano (1995).

Full references provided in bibliography slide.

Model Justification

Existing literature supports both the economic drivers of shipping demand and the use of regression-based methods to analyze and forecast container shipping activity. Existing literature has not examined the impacts of all four factors at the route level in the UK.

Data Overview

Category	Variable Name	Description	Unit	Frequency	Source	Role
Dependent variable	Container Volume (TEU)	Monthly container throughput by route-direction	TEU	Monthly	Bloomberg / Port data	Dependent variable
Price variable	Freight Price Index	Market freight rate indicator	Index	Monthly	Bloomberg	Independent variable
Cost variable	Bunker Fuel Price	Fuel cost for vessel operations	USD per barrel	Monthly	EIA / Bloomberg	Independent variable
Policy control	US Economic Policy Uncertainty Index	Measures economic policy uncertainty	Index	Monthly	policyuncertainty.com	Control variable
Structural control	Congestion Dummy	Indicates port congestion periods	0 or 1	Monthly	Constructed	Control variable
Macro control	EU Industrial Production Index	Proxy for EU economic activity	Index	Monthly	OECD / Eurostat	Control variable
Macro control	UK Industrial Production Index	Proxy for UK economic activity	Index	Monthly	ONS / OECD	Control variable
Trade control	Trade Imbalance	Export-import difference by route	Currency value	Monthly	ONS trade data	Control variable
Financial control	GBP/EUR Exchange Rate	Currency competitiveness measure	Ratio	Monthly	Bank of England	Control variable
Price-level control	UK CPI	Inflation indicator	Index	Monthly	ONS	Control variable

Panel characteristics:

- Geographic scope: UK–France, UK–Germany, UK–Netherlands
- Time period: 2016–2024
- Frequency: Monthly
- Unit of observation: Route × Direction × Month
- Structure: Balanced panel | Routes: 3 | Directions: 2
- Total observations: 648

Panel size: 648 observations (3 routes × 2 directions × 108 months)

Outcome variable definition:

Dependent variable: $\ln(\text{Container Volume (TEU)})$

Container volume is measured monthly at the route-direction level and log-transformed to stabilize variance and enable elasticity interpretation in the regression model.

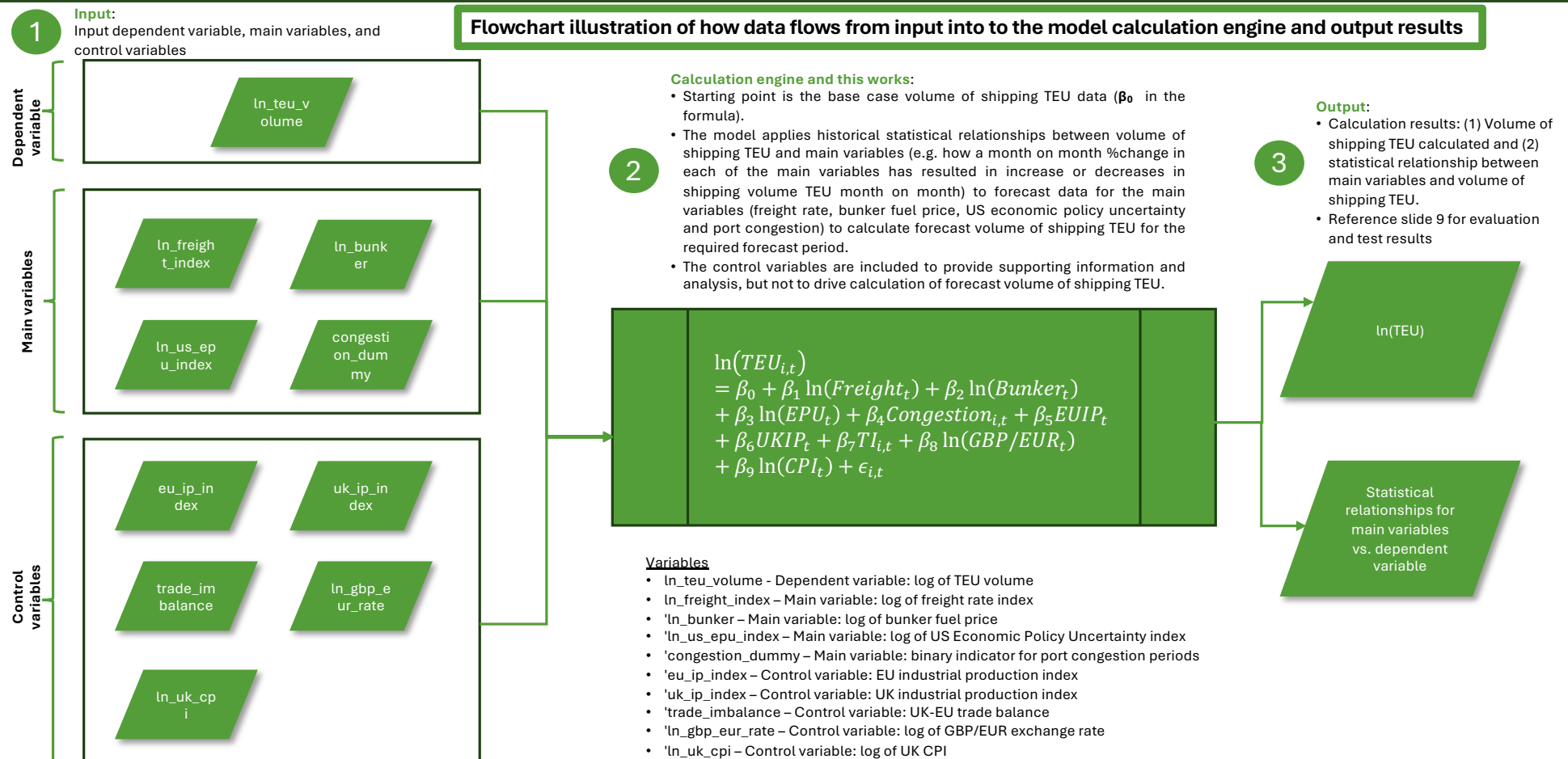
Interpretation:

Coefficients measure percentage change in shipping volume in response to percentage changes in explanatory variables.

Variables & Economic Intuition

Variables	Economic Rationale	Assigned Variable in Model Calculation for this Exercise
TEU volume	TEU is Twenty-foot Equivalent Unit. The twenty-foot-long intermodal container is the internationally recognized standard for measuring cargo capacity on a cargo ship.	Dependent variable (This is the output data the model is predicting or forecasting)
Freight rate	Higher freight rates increase transportation costs, and these are passed on to consumers, causing increased retail prices potentially decreased consumers demand and trade which reduces shipping volume (economics of demand and supply).	Main variable
Bunker fuel price	- Bunker fuel refers to any type of fuel that is used to power ships. Increasing bunker fuel prices results in surcharges, higher freight costs, and potentially lower shipping volumes. - A key point to highlight is that shorter shipping routes (e.g., UK to EU countries reviewed for this exercise) consume lower fuel per ton of goods and are less affected by rising bunker fuel prices and shipping companies may prioritize these shipping routes due to the cost effectiveness which would yield a positive price elasticity relationship contrarian to longer shipping routes which would have expected negative price elasticity impact from rising fuel costs.	
US economic policy uncertainty	Whilst historical data (2016 – 2024) analysed as part of this exercise does not indicate a strong relationship between US economic policies and shipping volume between UK and EU member states, we have decided to future proof the model by adding US economic policy uncertainty as a main variable input into the model calculations given recent US tariffs imposed on EU and UK (and other countries) in 2025 which could lead to deepening integration among EU member states and acceleration of initiatives to consolidate internal trade, harmonise regulations and promote intra-European supply chains.	
Port congestion periods	- Congestion in some of the busiest ports including Rotterdam, Hamburg, Antwerp has resulted in delays of up to 48 to 72 hours waiting time for container vessels to berth and this tends to peak during summer months and there are other factors contributing to this e.g., labor shortages, labor strikes etc. Port congestion is not a variable that will reduce shipping volume per se, but it could potentially lead to cargo vessel operators exploring alternative ports that are less busy. - Congestion has been added as a main variable to inform tactical planning of shipping routes in the short term.	
EU industrial production index	Shipping contracts are pre-agreed from a few weeks to several months in advance and are not affected by short term indicators such as industrial production index which may not materialise in expected GDP prediction. Hence, both the EU and UK indicators have been included as control variables which would not drive the forecast of volume of shipping volume TEU but will help provide a sign on the potential direction of the economy.	Control variable
UK industrial production index		
UK-EU trade imbalance		
GBP-EUR rate		
UK CPI		

Data Processing & EDA



Model

Two Stage Econometric Model

Structural Demand Estimation OLS Panel Regression

- Estimates route-specific elasticities from **2016-2024 panel data**
- Fits **6 separate models** (3 routes x Exports/Imports)
- Output coefficients highlight the link between **freight volume & independent variable**
- This model helps quantify **freight pricing, crisis, political turmoil, and bunker fuel impacts on shipping volume**
- The model was developed referencing academic literature Chen et al. (2017), Agnolucci et al (2015), Head & Mayer (2014), and Lin et al. (2023)
- We used heteroskedasticity- and autocorrelation-adjusted standard errors to address autocorrelation in the shipping data.
- We also included fixed effects to account for trading pair heterogeneity

$$\begin{aligned} \ln(TEU_{i,t}) &= \beta_0 + \beta_1 \ln(Freight_t) + \beta_2 \ln(Bunker_t) + \beta_3 \ln(EPU_t) \\ &+ \beta_4 Congestion_{i,t} + \beta_5 EUIP_t + \beta_6 UKIP_t + \beta_7 TI_{i,t} + \beta_8 \ln(GBP/EUR_t) \\ &+ \beta_9 \ln(CPI_t) + \epsilon_{i,t} \end{aligned}$$

(*TI*): trade imbalance where (*i*): route
– direction pair, (*t*): month

Out of Sample Forecasting OLS

What is it doing?

Model 1: $\widehat{y_{2023-24}} \sim \text{Train}(2016 - 2022)$

Model 2: $\widehat{y_{2024}} \sim \text{Train}(2016 - 2023)$

- Referencing Diebold 1995 methodology, we used the 2016-2023 dataset to train our model
- We are looking at the MAPE, MAE, RMSE & R² of our test results.
- Model 1 was done for validation

Model 2 Formula:

$$\begin{aligned} \widehat{y_{i,2024}} &= \widehat{\beta_0} + \widehat{\beta_1} \ln(Freight_{2024}) + \widehat{\beta_2} \ln(Bunker_{2024}) + \widehat{\beta_3} \ln(EPU_{2024}) \\ &+ \widehat{\beta_4} Congestion_{i,2024} + \widehat{\beta_5} EUIP_{2024} + \widehat{\beta_6} UKIP_{2024} + \widehat{\beta_7} TI_{i,2024} \\ &+ \widehat{\beta_8} \ln(GBP/EUR_{2024}) + \widehat{\beta_9} \ln(CPI_{2024}) \end{aligned}$$

Evaluation & Results

Full Levels Model Results						
	Exports 	Imports 	Exports 	Imports 	Exports 	Imports 
Freight Rate	-0.1158***	-0.1156***	-0.0713***	-0.0739***	-0.0941***	-0.0676***
Bunker Fuel Price	0.3821***	0.3745***	0.3392***	0.3026***	0.1647**	0.1417**
US Policy Uncertainty	0.0090	0.0063	0.0109	0.0187	-0.0385	-0.0272
Congestion	0.0351	0.0358	-0.0557*	-0.0485	0.0074	0.0043
R ²	0.539	0.769	0.337	0.653	0.746	0.813

- **Freight rate elasticity** is negative and statistically significant across all routes, with coefficients ranging from **-0.07 to -0.12**
- **Bunker fuel price** elasticity is **positive and highly significant** for all routes, strongest for France and the Netherlands
- **Congestion** (at respective ports) becomes **insignificant** in most full models, except weakly negative for the UK-Netherlands
- **US policy uncertainty** is insignificant in all specifications.
- **Model fit (R²)** is strongest for UK-Netherlands and UK-France imports, weaker for Germany exports
- Forecasting accuracy is moderate overall, with the strongest for Germany (**11-14% MAPE**) and the Netherlands (**13%**), indicating greater volatility in the UK-France corridor

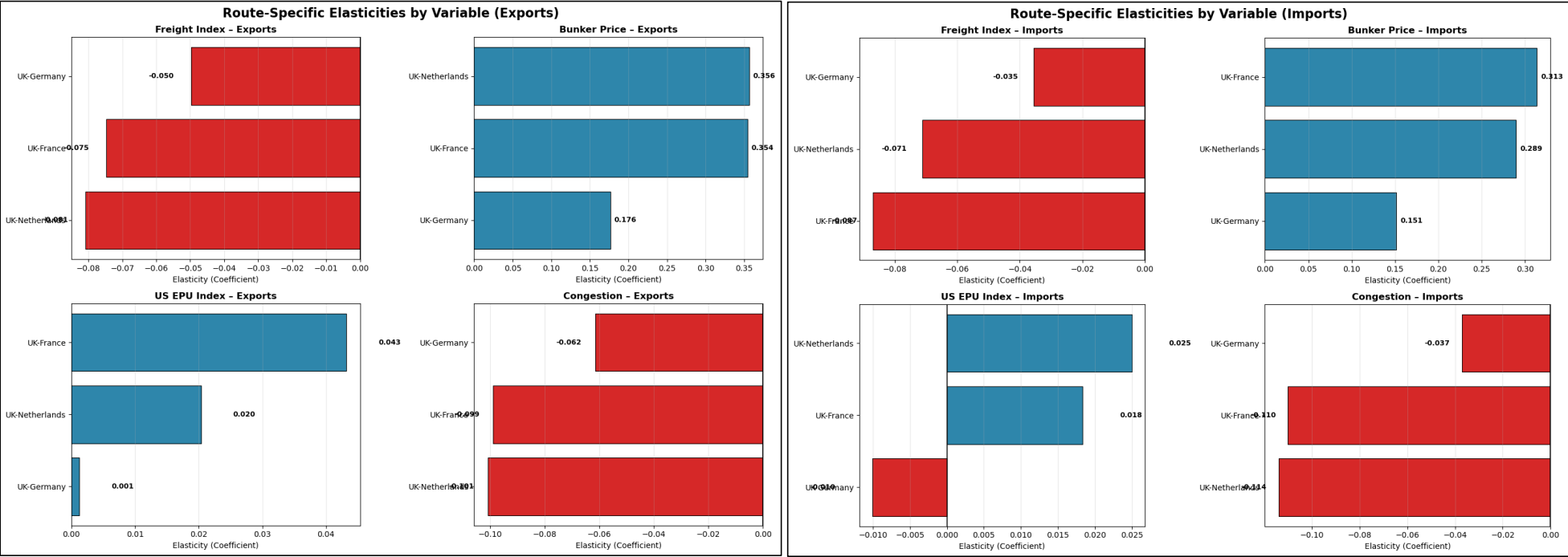
Model Back Test Based on 2024 Forecast Performance				
Route	Direction	RMSE (ln)	MAPE (%)	R ²
	Exports	0.2710	20.61	-1.591
	Imports	0.2629	20.56	0.198
	Exports	0.2013	14.27	-0.478
	Imports	0.1545	11.01	-2.734
	Exports	0.1677	13.50	-0.595
	Imports	0.1804	13.53	-1.881

- Back test results highlight moderate model forecasting accuracy. The intention is that the model should be used in parallel with senior management overlay and expert judgement for decision-making

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$

Note: the model was back tested for 2022 and 2023 and as expected the accuracy was low due to major geo-political events e.g. Ukraine invasion (2022) and war in the Middle East (2023). The 2024 back test result is a better indication of the model capability and has been presented above in the results.

Evaluation & Results

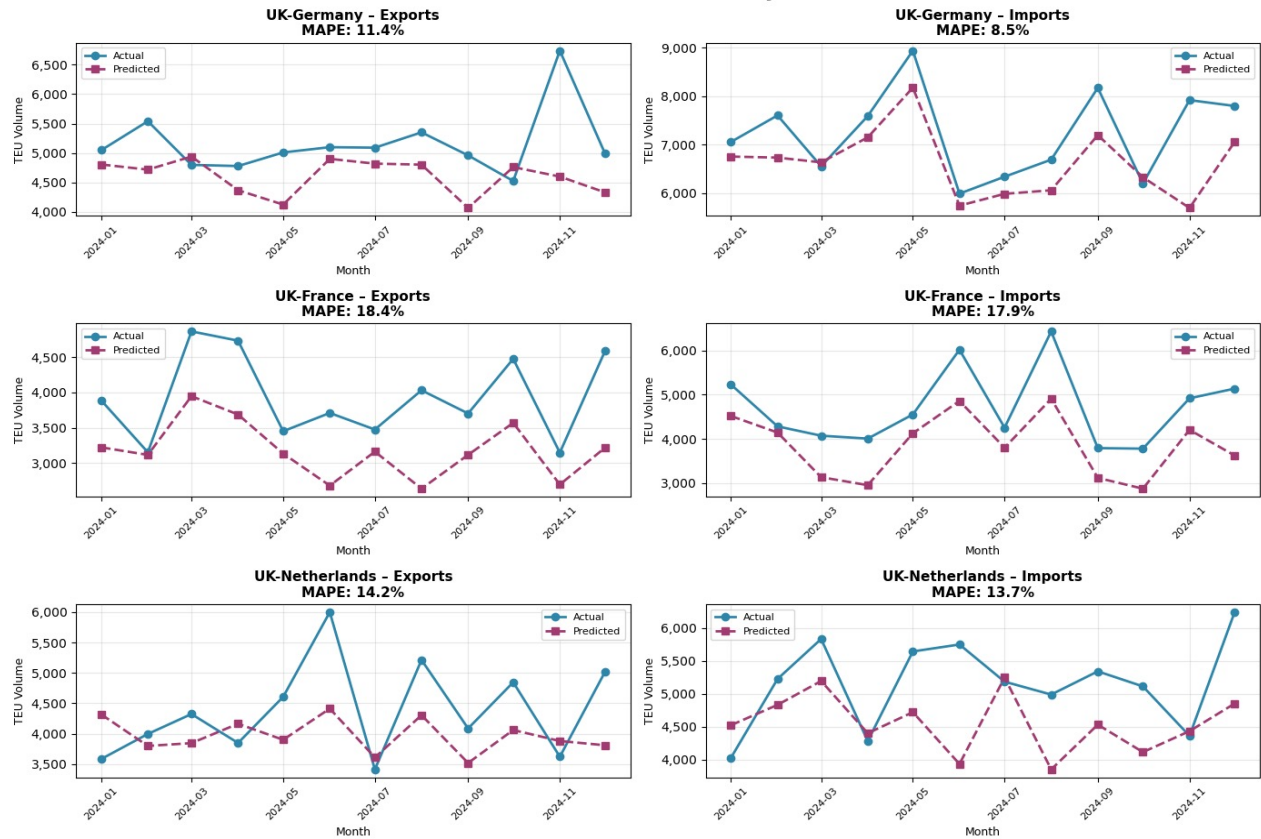


Route Specific Price Elasticities (Exports)

Route Specific Price Elasticities (Imports)

Evaluation & Results

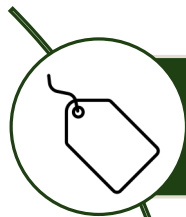
2024 TEU Volume: Actual vs Predicted by Route-Direction



Brief Description

- Forecast Accuracy is Strongest for Germany
- The Netherlands shows moderate and stable predictive performance
- France shows the largest deviations, indicating greater volatility

Insights & Implications



Freight Costs Significantly Reduce Volumes Across All Routes



Political Risk and Congestion have no statistically significant effect on the models. These have been included to future-proof the model, given recent US foreign policy turmoil & recent congestion issues across global ports.

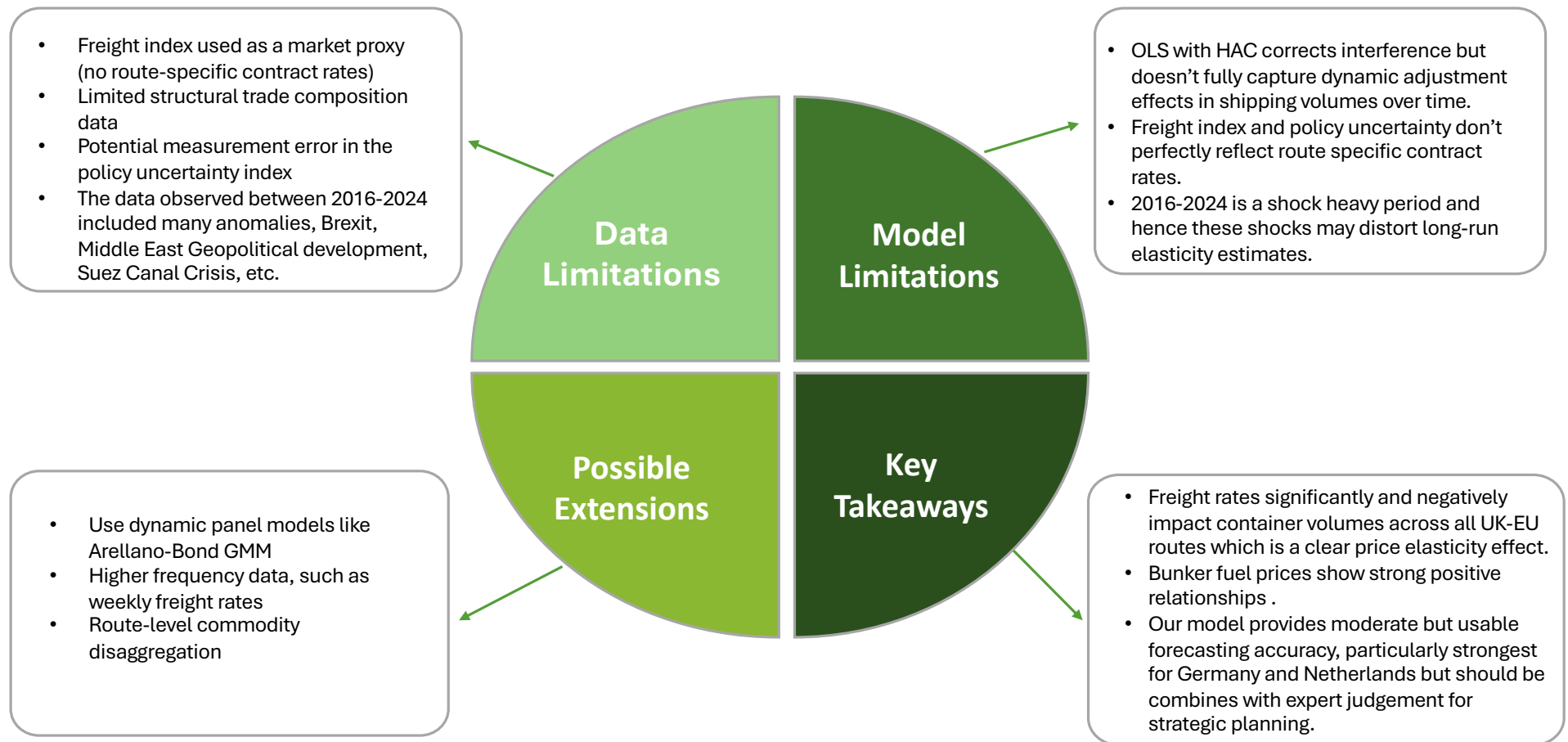


Higher Fuel Prices Reallocate Trade Toward Shorter Routes



Forecasting Performance Varies Sharply by Corridor

Limitations & Conclusion



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Correlation Matrix

